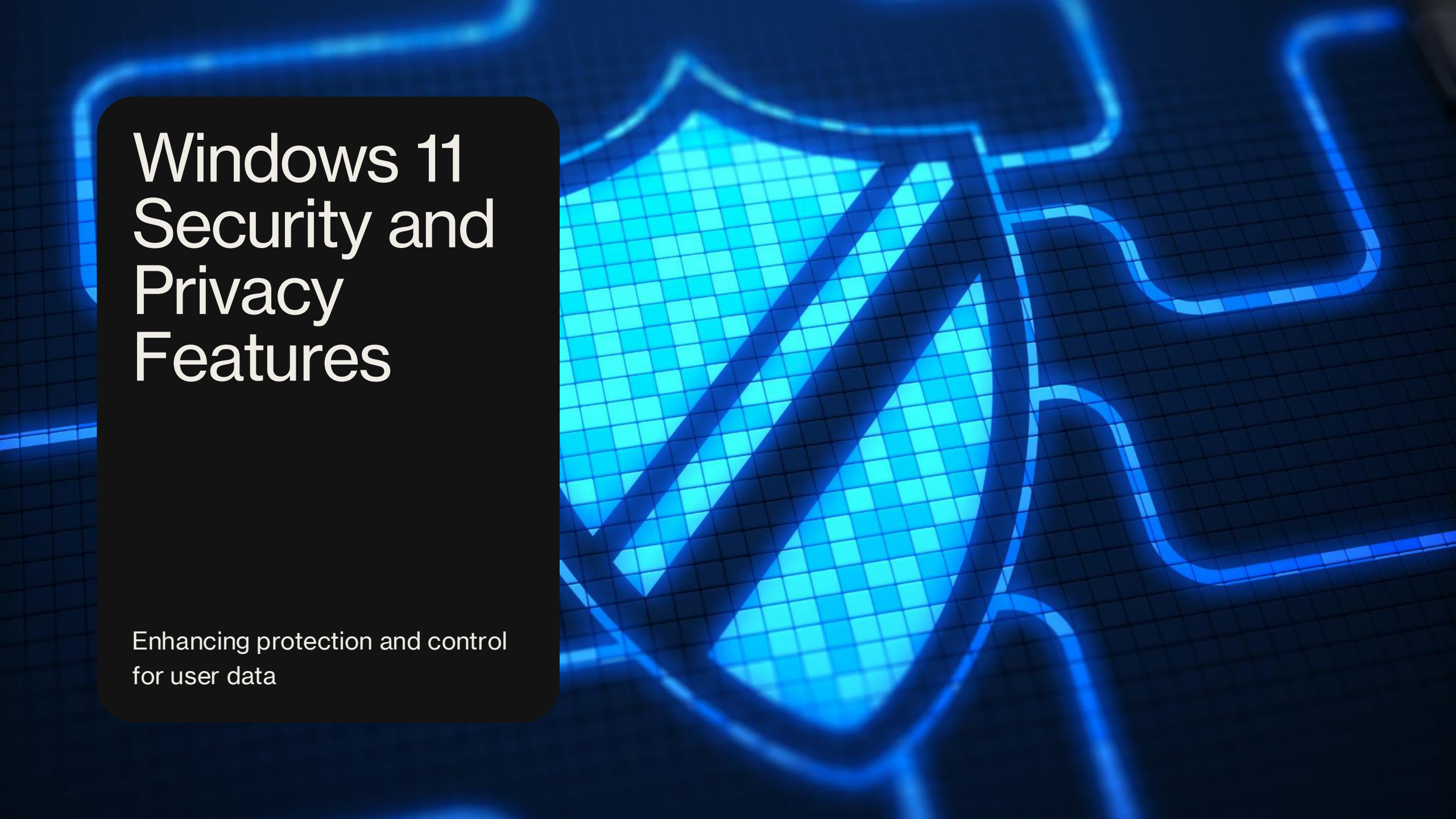
Windows 11 Security and Privacy Features

Enhancing protection and control for user data



Windows 11 Security and Privacy Features

User Account Control (UAC)

Security Feature Purpose

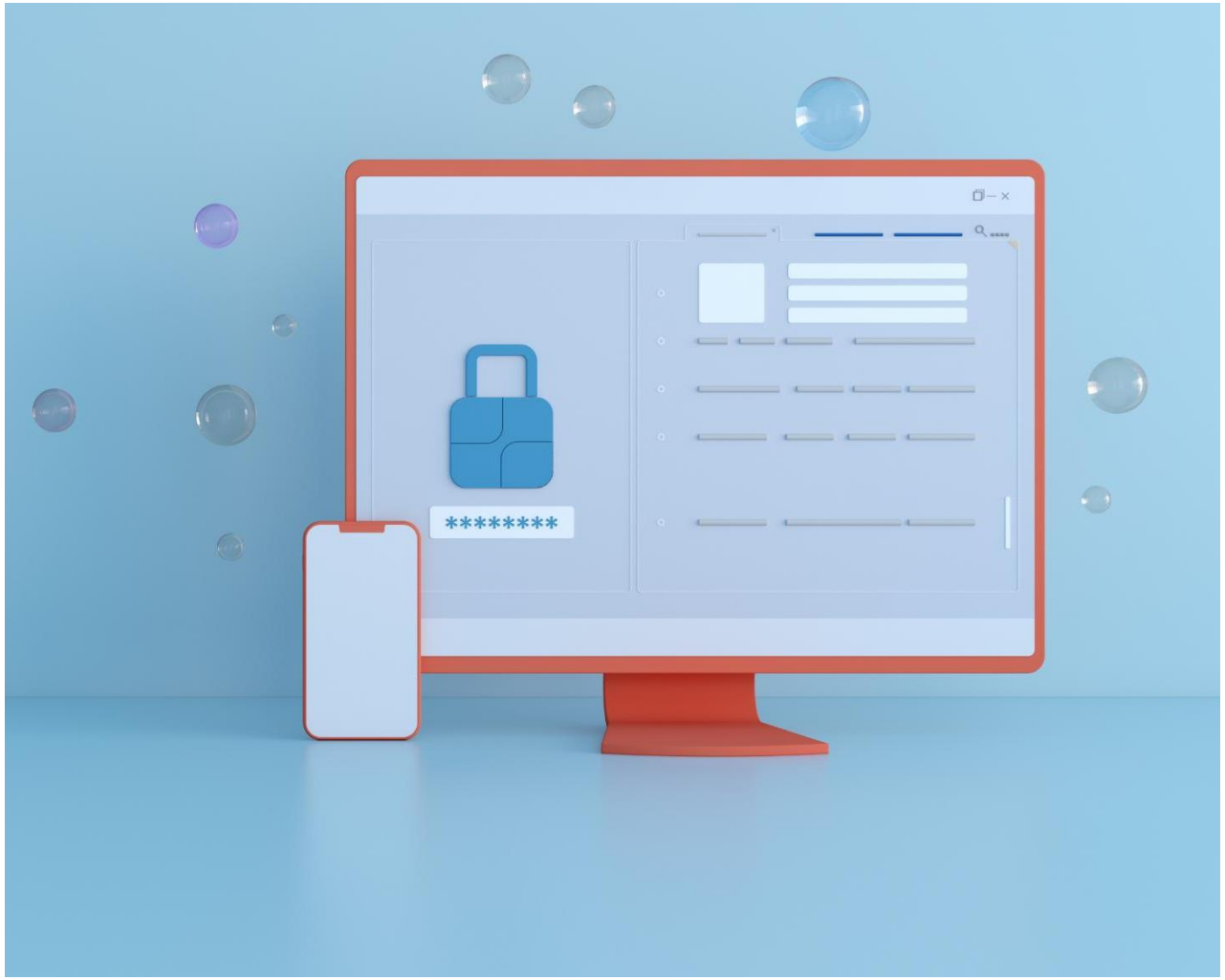
UAC prevents unauthorized changes by requiring user permission or administrator password for important actions.

Malware Impact Mitigation

UAC helps mitigate malware impact by blocking unauthorized software installations and accidental system changes.

Configurable Strictness Levels

Users and organizations can configure UAC strictness to balance system security with usability.



Microsoft Defender SmartScreen

Cloud-Based Threat Protection

SmartScreen uses a cloud database to continuously check files and web content against known threats.

File Reputation Evaluation

Unrecognized files are evaluated for reputation, warning users about potentially unsafe downloads.

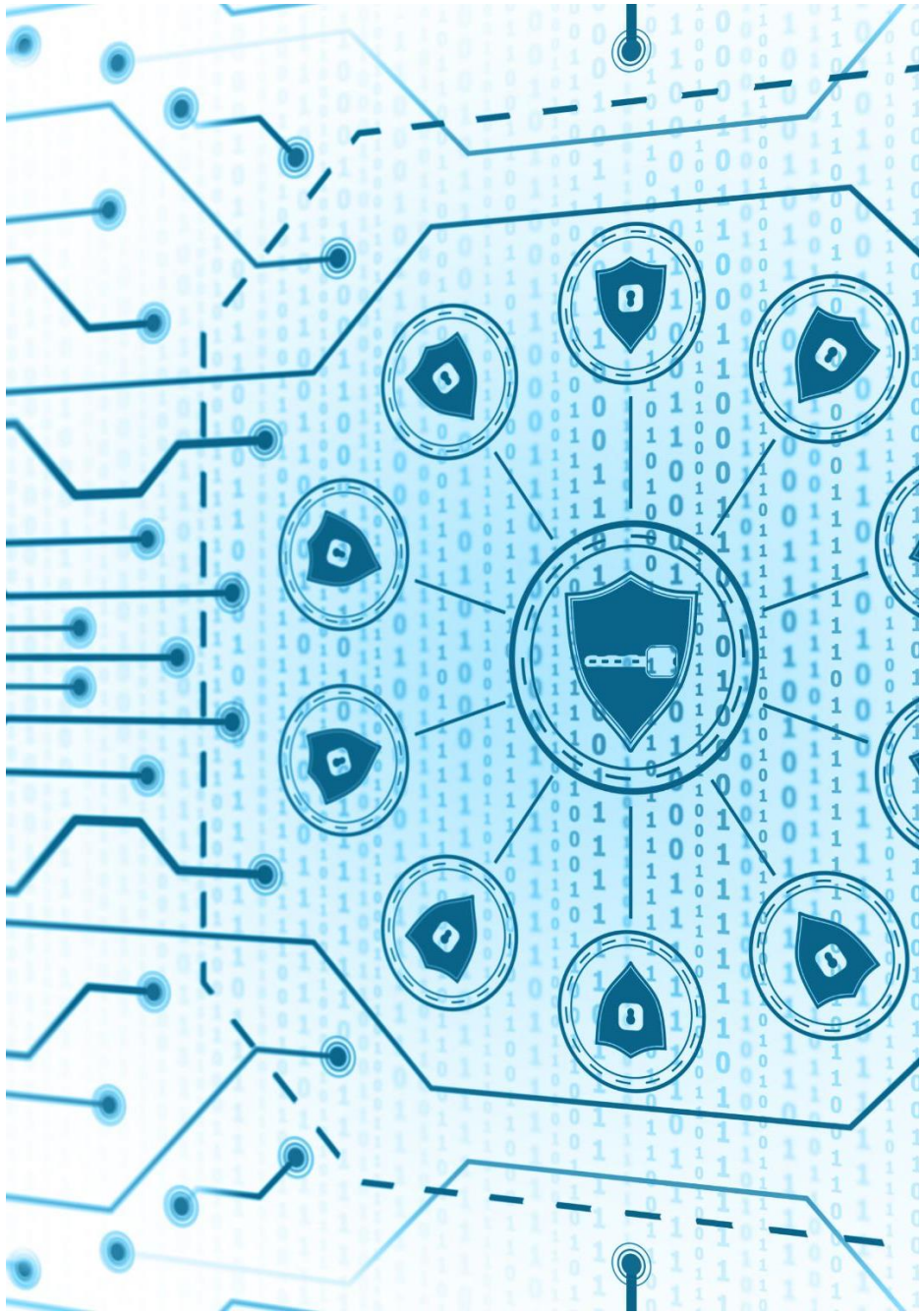
Integration with Windows

SmartScreen integrates with Microsoft Edge and Windows for real-time protection against malicious websites and software.

Phishing and Social Engineering Defense

The feature helps prevent phishing and social engineering attacks by alerting users before running harmful software.





Windows Defender Firewall

Traffic Monitoring and Control

The firewall monitors and controls network traffic based on security rules to block unauthorized access.

Barrier Between Device and Internet

It acts as a security barrier between the user's device and the internet to protect against threats.

Customizable Network Profiles

Users can configure firewall settings for private, public, or domain networks to tailor protection.

Application and Port Rules

Custom rules allow or block specific applications and ports to enhance network security.

BitLocker Drive Encryption

Full-Disk Encryption

BitLocker encrypts entire drives to protect data from unauthorized access on lost or stolen devices.

Secure Key Storage

Uses Trusted Platform Module or USB key to securely store encryption keys for authorized access.

Management and Recovery

Managed through Control Panel or Settings with recovery keys to regain access after hardware changes.



Microsoft Defender Antivirus

Real-Time Protection

Microsoft Defender Antivirus continuously scans files and applications to prevent malware infections in real time.

Automatic Updates

The antivirus regularly updates threat definitions to defend against the latest viruses and malware.

Cloud-Delivered Protection

Cloud-based machine learning quickly detects emerging threats using big data analysis and AI.

Built-in Windows Security

Integrated into Windows 11, it provides baseline protection without needing third-party antivirus software.



Windows Update and Patch Management

Security and Stability

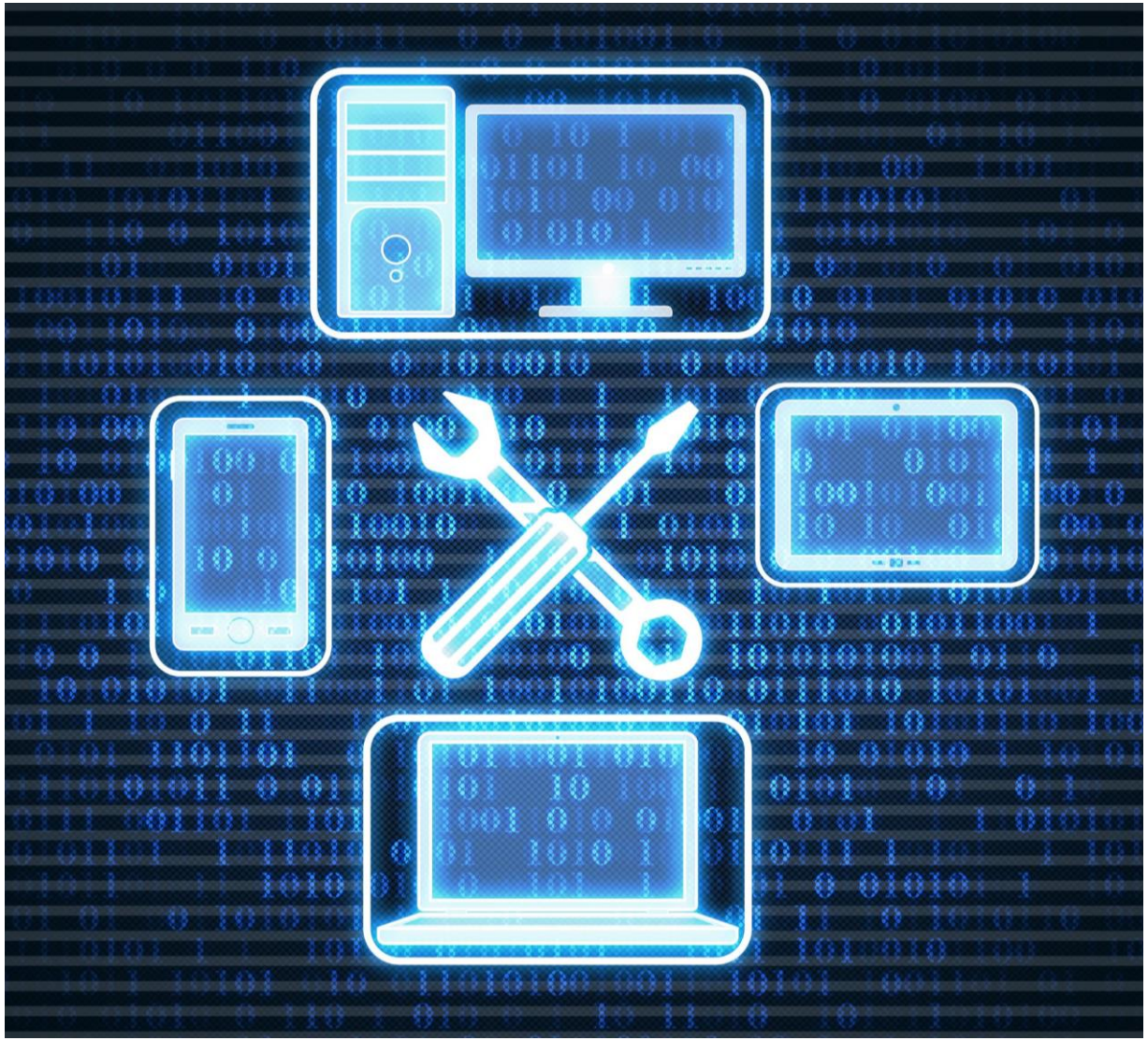
Regular Windows updates fix vulnerabilities and improve system stability, enhancing device security and performance.

User Update Configuration

Users can choose automatic or manual update installation based on their preferences for system control.

Centralized Management Tools

Organizations use tools like WSUS and Microsoft Endpoint Manager to deploy and control updates efficiently.



Phishing Awareness and SmartScreen Alerts

Phishing Attack Awareness

Users must learn to identify suspicious emails with poor grammar and urgent requests to avoid phishing traps.

SmartScreen Protection

SmartScreen warns users before running unrecognized or potentially malicious applications, enhancing security.

Combining Awareness and Security

User education combined with built-in security features reduces the risk of phishing scams effectively.



Privacy Settings and Data Collection

Comprehensive Privacy Controls

Windows 11 provides extensive privacy s
to control data collection and sharing w
Microsoft.

Privacy Dashboard Features

The Privacy Dashboard enables users t
manage permissions for location, camer
microphone, and other system features

User Empowerment and Transparenc

Windows 11 emphasizes transparency and
empowers users to make informed decisions
about their data privacy.





Controlled Folder Access

Protection Against Ransomware

Controlled Folder Access blocks unauthorized applications from making changes to protected folders, preventing ransomware attacks.

Trusted Application Access

Only trusted applications specified by the user can access protected folders, ensuring data integrity and security.

User-Specified Folder Protection

Users can designate which folders to protect, tailoring security to critical data and organizational needs.

Enhanced Data Security

This feature strengthens defense against data loss and corruption in business and academic environments.

Exploit and Tamper Protection

Exploit Protection Features

Provides system-wide and application-specific defenses against known cyberattack techniques.

Tamper Protection Mechanism

Prevents unauthorized changes to security settings, maintaining antivirus and firewall integrity.

Integrated Security Defense

Combines exploit and tamper protections to ensure robust system security against advanced threats.



Windows Hello and Local Security Authority

Windows Hello Features

Windows Hello uses facial recognition, fingerprint scanning, and secure PINs for fast, secure sign-ins.

Local Security Authority Role

The Local Security Authority enforces security policies and manages user authentication in Windows.

Enhanced Identity Protection

Windows Hello and LSA work together to ensure only authorized users access system resources securely.



Summary

Core Security Features

Windows 11 offers User Account Control, SmartScreen, Defender Firewall, BitLocker, and Microsoft Defender Antivirus to protect users.

Managing Threats and Updates

Effective management of updates, phishing recognition, and privacy configuration enhances system safety and user awareness.

Advanced Protections

Controlled Folder Access, Exploit Protection, and Tamper Protection help safeguard data and system integrity on Windows 11.

Identity Protection

Windows Hello and Local Security Authority provide robust identity verification to secure user access.

